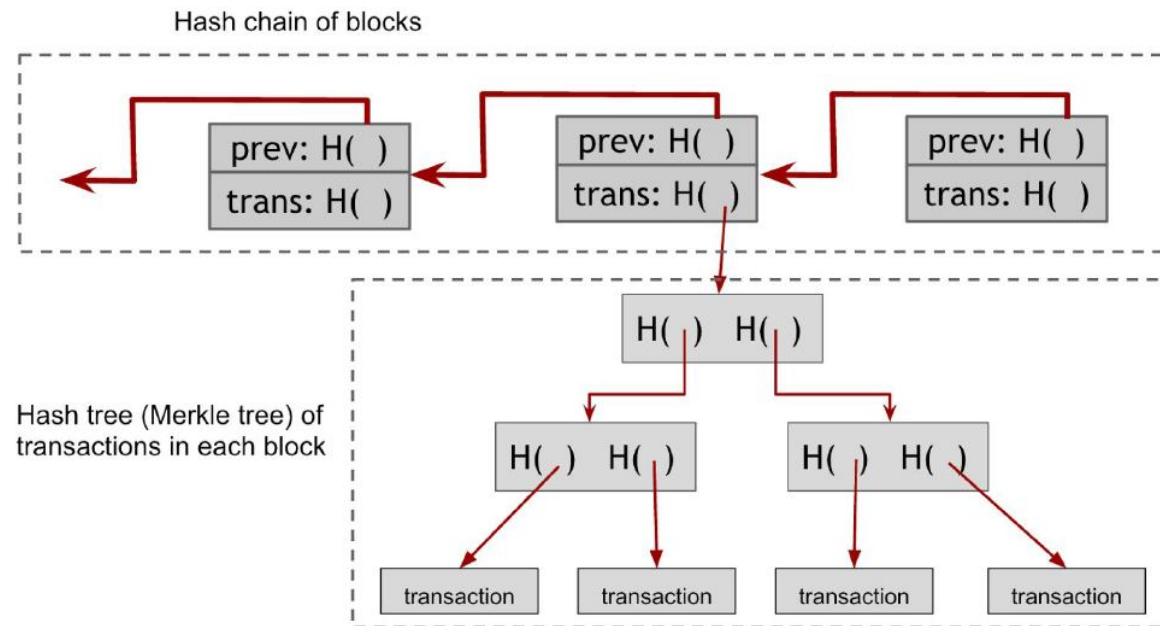


Bitcoin Blocks:



If you need to verify a specific transaction in a block, you only need the hashes along the path to the Merkle root, rather than the entire transaction list.

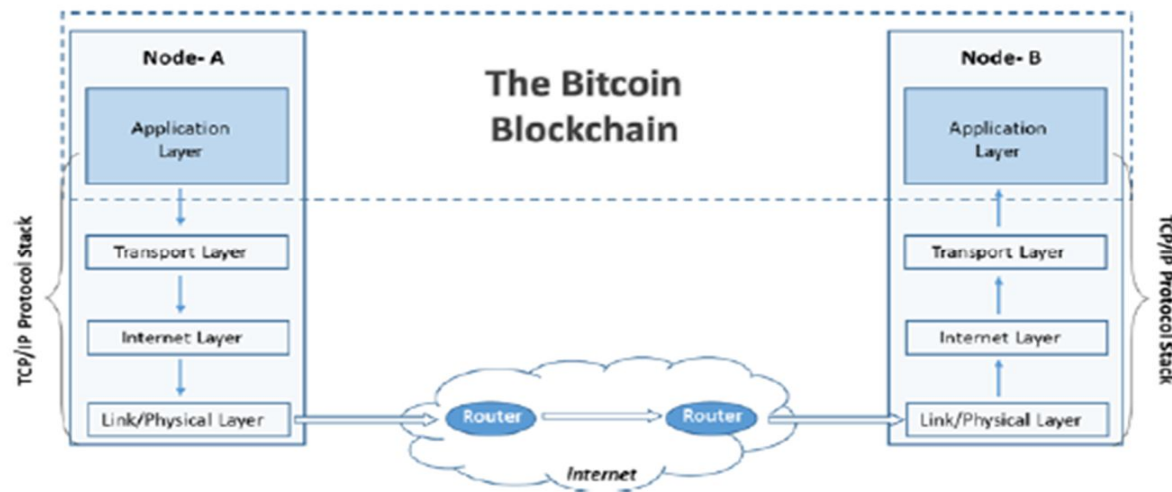
- Coinbase Transactions:

```
"in":[
  {
    "prev_out":{
      "hash":"000000.....0000000",
      "n":4294967295
    },
    "coinbase":"..."
  },
]
"out":[
  {
    "value":"25.03371419",
    "scriptPubKey":"OPDUP OPHASH160 ... "
  }
]
```

Historical Context: The first Bitcoin block (Genesis Block) included a message from the London Times about bank bailouts, symbolizing Bitcoin's decentralized ethos.

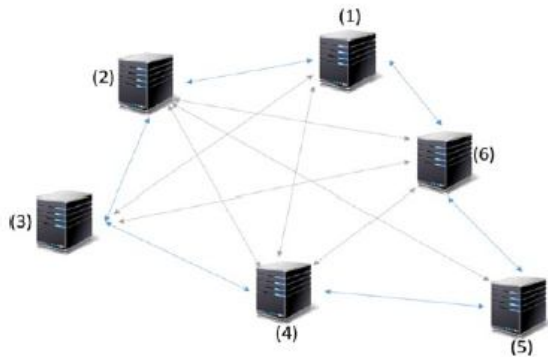
The Bitcoin Network:

The Bitcoin network is a peer-to-peer (P2P) system that ensures the decentralized propagation of transactions and blocks. It operates without any central authority, relying on the cooperation of nodes.

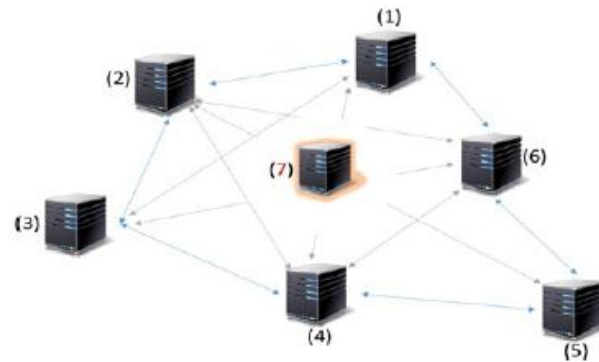


If Alice attempts to double-spend by broadcasting conflicting transactions, nodes temporarily disagree on which transaction to include. The conflict is resolved once a block is mined.

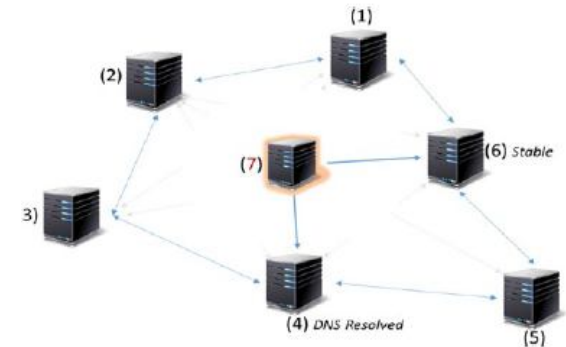
Network discovery for a new node:



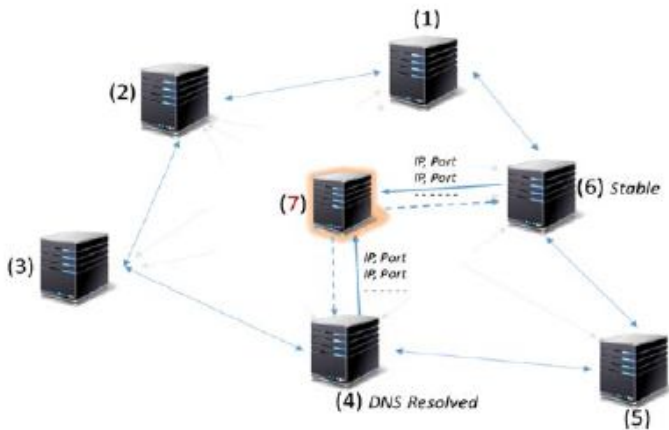
Step 1: Existing Bitcoin Network



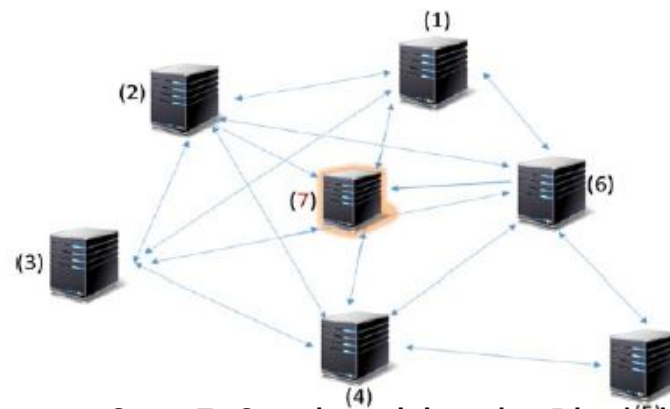
Step 2: A new node attempting to connect to the network.



Step 3: The new node querying DNS seeds and contacting a peer



Step 4: Peer Discovery and



Step 5: Synchronizing the Blockchain: The new node syncing with

- Full nodes, Lightweight nodes
- Limitations and Improvements of Bitcoin protocol

1 .Throughput Issues

- **Block Size Limit:** 1MB per block.
- **Result:** Bitcoin can handle only 7 transactions per second (TPS).
- **Comparison:**
 - Visa processes ~2,000 TPS on average and peaks at ~10,000 TPS.
 - PayPal can handle ~100 TPS.

Impact:

Slow transaction processing leads to delays and higher fees during peak usage

2 Cryptographic Vulnerabilities

- Bitcoin relies on:
 - **SHA-1 and SHA-256** (hashing algorithms).
 - **ECDSA** (signature algorithm using secp256k1 curve).
- Risks:
 - Advances in cryptanalysis could render these algorithms insecure.
 - Upgrading cryptographic primitives would require consensus and potentially a hard fork.

3 Protocol Constraints

- Hard-coded limits include:
 - Total Bitcoin supply: 21 million coins.
 - Mining reward halving every 210,000 blocks (~4 years).
- These constraints are unlikely to change due to their economic implications.

4. Network Efficiency

- The flooding protocol introduces significant latency:
 - Blocks take up to 30 seconds to propagate to all nodes.
 - Bandwidth bottlenecks slow down large blocks.

5 Governance Challenges

- Upgrading Bitcoin's protocol is difficult due to the decentralized nature of the network.
- **Hard Forks:**
 - Introduce backward-incompatible changes.
 - Split the blockchain into separate versions.
 - Example: Bitcoin Cash split from Bitcoin in 2017 over block size disagreements.
- **Soft Forks:**
 - Introduce stricter rules without breaking compatibility.
 - Example: Segregated Witness (SegWit) reduced transaction sizes by modifying the data structure.

- **Conclusion**

- Bitcoin's block and network architecture ensure decentralization and security but face significant challenges in scalability and governance.
- Potential solutions, like the Lightning Network and other Layer 2 technologies, aim to improve throughput while preserving Bitcoin's core principles.